

- Document Control
- System Name: Application
- Version:
- Prepared By:
- Date:
- Status:

- Executive Summary
- The document outlines business requirements for Application to process, cleanse, aggregate, and manage customer and order data sourced from CSV files, using persistent delta tables and automated workflows.

- Business Objectives
- Efficiently ingest customer and order data from predefined volumes.
- Clean data by removing nulls and duplicates.
- Join customer and order datasets for unified views and history.
- Aggregate total spend at customer and date level.
- Automate workflow with delta live tables.

- In-Scope
- Loading of CSV data for customer and order tables.
- Schema definitions for all tables.
- Data cleansing (null and duplication removal).
- Computation of total amount in orders.
- Building and maintaining order summary with SCD type 2.
- Aggregating and storing customer spend.
- Workflow orchestration via delta live tables.

- Out-of-Scope
- Data ingestion from non-CSV sources.
- Any downstream application processing outside specified tables.
- Real-time event streaming and processing.
- Advanced analytics or reporting layers.

- Stakeholders
- Data Engineering Team
- Business Analysts
- Data Governance

- Business Requirements

BR-SDL-001 — Load Customer Data from CSV

- Description: Ingest customer CSV data from specified volume into persistent delta table.
- Business Value: Centralizes customer information for downstream consumption.
- Priority: High

BR-SDL-002 — Load Order Data from CSV

- Description: Ingest order CSV data from specified volume into persistent delta table.
- Business Value: Centralizes order information for unified analysis.
- Priority: High

BR-SDL-003 — Data Cleansing for All Tables

- Description: Remove null and duplicate records from both customer and order tables.
- Business Value: Improves trust and accuracy for analytics.
- Priority: High

BR-SDL-004 — Compute Total Amount in Orders

- Description: Add column "TotalAmount" to order table—computed as $\text{PricePerUnit} * \text{Qty}$.
- Business Value: Enables spend analysis and reporting.
- Priority: High

BR-SDL-005 — Create and Maintain Order Summary with SCD Type 2

- Description: Join customer and order tables by CustId; load to ordersummary as SCD type 2 table, maintaining history.
- Business Value: Maintains change history for compliance and auditing.
- Priority: High

BR-SDL-006 — Update Order Summary on Customer Changes (SCD Type 2 Logic)

- Description: Update ordersummary so old records become Inactive and new records Active; manage StartDate/EndDate for each version.
- Business Value: Ensures historical accuracy and lineage.
- Priority: High

BR-SDL-007 — Aggregate Customer Spend

- Description: Aggregate TotalAmount by Name and Date from ordersummary table, load to customeraggregatespend table.
- Business Value: Provides insights into customer purchase behaviour over time.

- Priority: Medium

BR-SDL-008 — Automate Workflow Using Delta Live Tables

- Description: Implement all ingestion, cleansing, join, and aggregation steps using delta live tables.
- Business Value: Orchestrates data pipeline for robustness and efficiency.
- Priority: High
- Assumptions & Constraints
 - All CSV files conform to the defined schema.
 - Volume paths are accessible by the system.
 - Delta lake infrastructure is available.
 - No upstream schema drift expected.
- Success Metrics
 - Data loads complete with zero errors.
 - All target tables are created and maintained in expected format and frequency.
 - Order summary table retains correct history on change events.
 - Aggregated spend records accurately reflect underlying data.
- Risks & Mitigations
 - Source files missing/incorrect format — Mitigate with pre-ingestion checks.
 - Table schema changes — Mitigate with schema enforcement and alerting.
 - Delta table write errors — Mitigate via error handling and retry logic.
- Approval
 - Business Owner:
 - Data Engineering Lead:
 - Date:
 - Status: